

Dawit Chun

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Education

Taejae University

B.S. in Data Science/AI & Natural Sciences (Physics)

Seoul, South Korea

Expected June 2027

- Matriculation Award: Admitted as youngest student (age 15) with highest entrance score
- Vice Student Body President, 2025

Publications

Peer-Reviewed Papers

1. **D. Chun**, M. Jun. "PDPM: A Morphable RISC-V Pipeline with Perceptron-Based Depth Prediction." *IEEE ICCD 2026*. **Accepted Aug 2026**.
2. D. Chun, **D. Chun**, Y. Cha, L. Jellibi, S.J. Park. "Watch-Out: Real-Time Tracking and Assistance for Wandering Alzheimer's Patients." *ACM CHI 2025 EA*. [doi:10.1145/3706599.3720308](https://doi.org/10.1145/3706599.3720308) **1st Place**.
3. **D. Chun**, D. Chun, W. Hwang. "AI-Driven Smart IoT Security Window Optimization System for Disaster Response." *J. Korea Society of Computer and Information*, 30(6), 2025. [doi:10.9708/jksoci.2025.30.06.147](https://doi.org/10.9708/jksoci.2025.30.06.147)
4. D. Chun, **D. Chun**, W. Hwang. "Smart Emergency Window System for Rapid Evacuation in Disasters." *J. Korea Society of Computer and Information*, 30(4), 2025. [doi:10.9708/jksoci.2025.30.04.093](https://doi.org/10.9708/jksoci.2025.30.04.093) **Excellence Paper Award**.
5. **D. Chun**, D. Chun, W. Hwang. "IoT-Implemented Environmental Hydroelectric Shower Module." *Korea Association for Science and Creativity*, 2023.

Pre-prints

1. **D. Chun**, Y. Cha, S.J. Park. "iPhoneme: Brain-to-Text Communication for ALS Using ConformerXL Decoding." *arXiv:2604.16441*, 2026. Equal contribution. [doi:10.48550/arXiv.2604.16441](https://doi.org/10.48550/arXiv.2604.16441)

Patents

1. **D. Chun**, D. Chun, W. Hwang. "AI-Based Integrated Smart Security Window and Automatic Control System." Korean Patent Application 10-2026-0029503, filed Feb 13, 2026.

Books (Co-author)

1. **D. Chun** et al. *Youth Conversations on Sustainable Development*. Office for Government Policy Coordination, 2025. Pub. No. 11-1092000-100007-01.
2. W. Hwang, **D. Chun**, D. Chun. *ChatGPT: Practical AI for the Classroom*. ISBN 9788970930084, 2024.

Research Experience

Research Intern

June 2026 – September 2026

ROBOTIS, Humanoid Software Team

- Benchmarking SOTA VLA and world models (GR00T N1.7, MolmoAct2, VLA-JEPA, FastWAM, LingBot-VA) on 5-DOF OMX, 6-DOF OMY, and the 22-DOF bimanual AI Worker for long-horizon manipulation tasks; published a YouTube walkthrough of OMY evaluations
- Designed and validated custom architecture variants—Prior-ACT (DINOv3-conditioned) and MoE-Diffusion policies. Prior-ACT improved success by 10% over Vanilla-ACT on long-horizon drawer and PnP placement manipulation tasks. Developed Gate-as-Potential, a potential-based shaping method for behavior cloning
- Tested human-in-the-loop learning (HG-Dagger, HIL-SERL via RLinf) and off-policy RL on Isaac Sim rollouts (TD3+BC, AWR, RECAP); built a teleoperation ROS2 system for human-in-the-loop RL with pause/resume across teleop, policy rollout, and auto-labeled interventions
- Extended Cyclo Intelligence, the in-house record/inference stack. Streamlined record-to-train-to-deploy path and documentations; bridged the ROBOTIS VLA environment with native LeRobot

Research Intern

Dec 2024 – Aug 2025

Ubion Corporation

- Redesigned autonomous vehicle hardware including PCB layout, servo motor integration, and chassis construction; migrated system from Raspberry Pi 4 to Pi 5
- Implemented deep learning models (YOLO, 2D CNN Regression) using TensorFlow, OpenCV, and Mediapipe for computer vision with variable-angle servo control
- Developed “DeepCo Board,” an ESP32-based AI-IoT education platform integrating on-board vision, audio, etc. sensors, custom PCB hardware, and mobile app deployment tools; demonstrated at COEX Education Fair 2025

STEM Instructor

July 2022 & 2023

KOICA / Korea Productivity Center, Cambodia

- Delivered robotics workshops to 30 high school teachers on Arduino programming, SolidWorks modeling, and C programming

Teaching Experience

Guest Instructor

July 2025

Geumcheon Science Cube Talent-Sharing Program, Seoul

4 hours

- Collaborated with Ubion’s DeepCo autonomous cars and Geumcheon Science Cube, supported by the Geumcheon local government.
- Proposed and delivered a hands-on lecture, “Learning AI through Self-Driving Robots,” to 20 elementary students (grades 4–6), guiding them through data collection, model training (LeNet-5), and the working principles of autonomous driving cars

Teaching Assistant

Fall 2025, Spring 2026

NS200: Physics in the Modern Society;

XD221: Quantitative Thinking in the Sciences, Taejue University

- Led weekly TA sessions; created and graded problem sets/projects; delivered advanced lectures beyond curriculum; connecting theoretical AI/electronics principles with math and physics

Fellowships

DYROS Robotics Bootcamp

August 2025

Seoul National University, Professor Hyeongki Lee & Jaehung Park

- Selected into intensive robotics program; developed control algorithms, motion planning, and learned ROS frameworks, inverse kinematics, and actuator control

Oxford-Stanford-Taejue Fellowship

Mar & Aug 2025

Kellogg College, Oxford & Taejue University, Seoul

Full funding (\$4,000)

- Selected as 1 of 8 Taejue representatives for AI governance and climate policy seminar at Oxford
- Won 1st place in AI Policy Debate in Seoul; donated \$2,000 prize to WWF; presented our work to the Deputy Prime Minister, contributing to discussions shaping Korea’s future AI policy

KAIST AI and Data Science Masterclass

Mar 2023 – Jan 2024

KAIST Center for Gifted Education

Grade: A+

- Year-long program; earned an A+ and was selected for the intensive DICIA-KAIST AI Master Class at KAIST (July 2023) on computer vision and algorithmic thinking

Honors & Awards

Talent Award of Korea, Ministry of Education

2025

Top 50 nationwide in University/Young Adult category (0.0002%). Awarded 2 million won; donated entire prize to Geumcheon Future Scholarship Foundation.

1st Place, ACM CHI 2025 Student Design Competition, Yokohama 2025
1st Place, Oxford-Stanford-Taejae AI Sustainability Policy Debate 2025
Gold Medal & Silver Medal, International Robot Olympiad, Autonomous Driving 2025, 2024
Excellence Paper Award, KSCI Winter Conference 2024
Gold Prize & 2nd Place, Aramco World Coding Cup; Hackathon Competition 2023
Gold Prize, KSME National Student Design & Engineering Competition 2022
 Additional: 5 × IRO medals (2023–25), Bronze KSME (2023), Geumcheon Hackathon 1st Place (2022), KAIST President’s Award (2021)

Skills

Programming: Python, C, VHDL, Verilog, SystemVerilog, Arduino, ESP32, Raspberry Pi
Tools: PyTorch, TensorFlow, OpenCV, ROS 2, MuJoCo, Isaac Lab, LeRobot, W&B, SolidWorks, 3D printing, Git, LaTeX, Jupyter
Domains: VLA models, imitation & reinforcement learning, world models, robotics, AI/ML, CPU architecture, autonomous systems, IoT

Certifications

• GPA: 4.45/4.5 • GRE: 326 (V162, Q164) • TOEFL: 118/120 • HSK Level 4 •
 OPIc: Advanced Low • Generative AI Expert Level 2

Volunteer Work

770 total hours, U9 Change translation services (272 hrs); Hosanna Church meal delivery (195 hrs); Gangnam-gu amphibian conservation (69 hrs); Geumcheon Science Classroom support (42 hrs); Cambodia STEM Teachers Training (192 hrs).

References

Sung Jun Park, Ph.D.

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